

Title: Battery SOC Estimation and Remaining Charging Time Prediction using ML

NAME:- SREEJA SANYAL

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Problem:

Accurate estimation of battery SOC and remaining charging time is important for battery monitoring and EV charging management. These quantities depend on nonlinear relationships among voltage, current, power, temperature, time and charging state.

Research Question:

Can regression models based on machine learning accurately estimate the state of charge and predict the remaining charging time from measurements taken during battery charging, while at the same time ensuring that the evaluation is free from leakage and reproducible?

Objectives:

- Inspect and understand the supplied charging dataset.
- Identify reliable input variables and targets.
- Build baseline regression models.
- Compare Random Forest, Linear Regression, Decision Tree & Gradient Boosting/XGBoost with simple baselines.
- Evaluate SOC and RCT separately using MAE, RMSE, MSE and R^2 .
- Analyze errors and identify important features.
- Produce a reproducible final pipeline.

Literature Findings and Research Framing

What the Literature Indicates:

- ML and DL methods are effective for data-driven battery SOC estimation.
- Voltage, current and temperature are important features for accurate SOC estimation.
- Neural-network models such as CNN, RNN, and LSTM show strong potential for SOC prediction.
- Model accuracy and robustness depend on dataset quality and varying battery operating conditions.
- Remaining charging-time prediction requires consideration of charging profiles and changing battery conditions.

Research Gap / Project Focus:

Instead of evaluating only one algorithm, this project will establish a common preprocessing and evaluation protocol and compare simple and advanced regression models for both SOC and RCT.

Main Risk:

Randomly splitting the data row-wise may put data from the same charging session into both training and testing. So, if a session or cycles is available, we will split the data based on sessions or cycles instead.

Research Framing & Mentor Review

1.

Battery Dataset



Data Understanding & Exploration



Data Cleaning



Feature Selection



Data Preprocessing



Train/Validation/Test Split



ML Model Training



SOC Estimation



Remaining Charging

Time



Model Evaluation

& Comparison



Best Performing Model

2.

Input Features:-

- Time/historical charging information
- Battery Voltage
- Charging Current
- Battery Temperature
- Charging power

Targets:-

- State of Charge (SOC)
- Remaining Charging Time (RCT)

Evaluation Metrics:-

- MAE
- RMSE
- R^2
- MSE

3.

Points to Confirm with Mentor:-

- Charging endpoint
- Dataset split strategy
- Availability of session/cycle identifiers
- Whether ANN is required
- Final preprocessing strategy

Current Assumptions:-

- SOC will be treated as a regression target.
- RCT will be treated as a regression target.